UNIVERSITY OF SCIENCE AND TECHNOLOGY OF HANOI



Application Of Tile Transposition Ciphers In Anti-Scraping Image Protection System

Full Name	Student ID	Major
Nguyen Tung Duong	23BA14076	ICT
Tran Mai Anh	2410093	ICT
Pham Dinh Anh Duong	23BA14075	ICT
Ta Quang Huy	23BA14037	ICT
Ha Dang Huy	23BA14138	ICT
Nguyen Minh Duc	23BA14056	ICT
Nguyen Minh Luong	23BA14184	ICT

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Contents

1 Introduction & Project Motivation **2**
1.1 Problem Statement: Automated Image Scraping and Economic Threats 2
1.2 Project Objectives 2

2 Theoretical Background **2**
2.1 From SRT to GRT 2
2.2 Image decryption via Inverse GRT (IGRT) 3
2.3 Key Terminology 3

1 Introduction & Project Motivation

1.1 Problem Statement: Automated Image Scraping and Economic Threats

Images are widely stored and distributed via websites and Content Delivery Networks (CDNs), which enable images to be loaded and collected automatically at low processing cost. This may allow unauthorized and illegal redistribution of images. In addition, existing image-protection methods may involve a trade-off between protection and performance. In pixel-chaotic ciphers, shuffling individual pixels across the entire image destroys spatial correlation, leading to an increase in compressed file size. Therefore, websites need to balance image protection, bandwidth costs, and display performance.

1.2 Project Objectives

- Reproduce and implement GRT-based tile-level image scrambling on the server (Python) and client-side viewport reconstruction (JavaScript Canvas).
- Validate the reversibility of GRT and confirm pixel-perfect image recovery with no quality loss.
- Measure computational latency, memory usage, and compression overhead across standard web formats (PNG, WebP, JPEG) to assess its feasibility for web-based image protection.

2 Theoretical Background

2.1 From SRT to GRT

Special Rectangular Transform (SRT) is a transformation method used to rearrange pixel positions, allowing accurate image reconstruction through its corresponding inverse transformation (ISRT) [1]. The SRT matrix is expressed as:

$$R_{\text{SRT}} = \begin{bmatrix} 1 & r \\ s & rs+1 \end{bmatrix} \tag{1}$$

where r and s are two integer parameters that determine the transformation matrix. However, its transformation matrix is constrained because the upper-left element is fixed at 1, resulting in only two degrees of freedom. To address this limitation, we adopt the GRT to enhance the flexibility of the transformation matrix and introduces a translation offset (r_m, r_n) to displace the origin (Figure 2c).

Unlike SRT, GRT involves 4 parameters (a, b, c, d), providing 4 degrees of freedom and greater flexibility in the transformation matrix [1]:

$$\begin{bmatrix} x \\ y \end{bmatrix}_{n+1} = \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}_n + \begin{bmatrix} r_m \\ r_n \end{bmatrix} \right\} \bmod \begin{bmatrix} M \\ N \end{bmatrix} \tag{2}$$

where, (a, b, c, d) are 4 transformation parameters, $G = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is a general 2×2 integer matrix whose determinant $\|G\| = ad - bc$ does not need to equal 1, $(x, y)_n$ are original pixel coordinates, $(x, y)_{n+1}$ are transformed tile coordinates, (r_m, r_n) are coordinate shifting offsets (Figure 2), and M, N are image height and width. However, the parameter set (a, b, c, d) mus satisfy these 4 mathematical conditions [1].

1. **Condition 1** — $(b \bmod h = 0)$ **or** $(c \bmod v = 0)$**.**
2. **Condition 2** — $\gcd(\|G\|, p) = 1$**.**
3. **Condition 3** — $\gcd(a, h) = 1$**.**
4. **Condition 4** — $\gcd(d, v) = 1$**.**

where $p = \gcd(M, N)$ is the greatest common divisor of the grid dimensions, and $h = M/p, v = N/p$ represent the irreducible aspect ratio. These conditions ensure a one-to-one mapping across rectangular grids without tile collisions.

2.2 Image decryption via Inverse GRT (IGRT)

To restore the original coordinates $(x, y)_n$ from a scrambled tile $(x, y)_{n+1}$, we apply the 4-stage Inverse GRT (IGRT) from the corollary [1] (Figure 1):

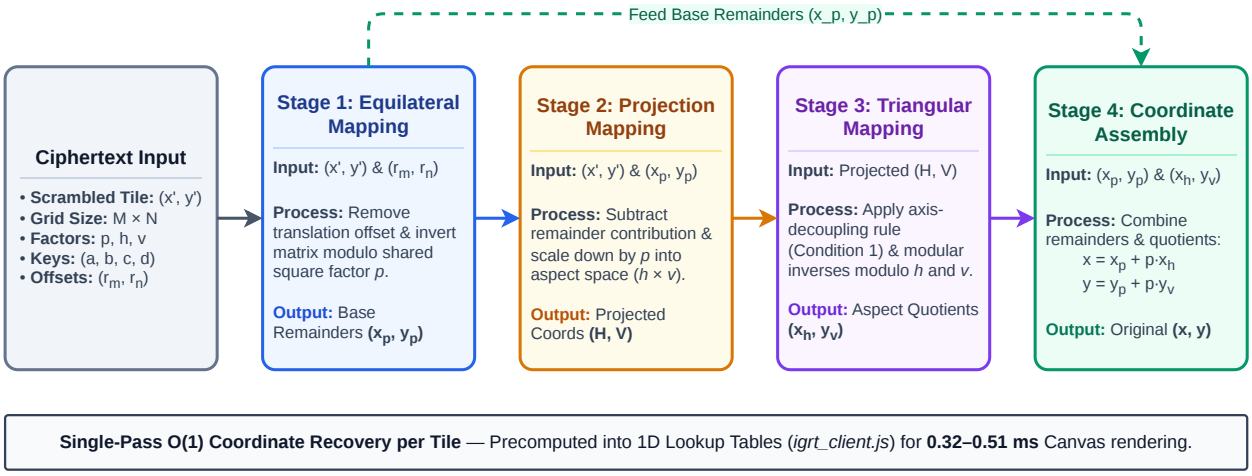


Figure 1: 4-Stage Single-Pass Coordinate Decryption in O(1) Time Complexity.

2.3 Key Terminology

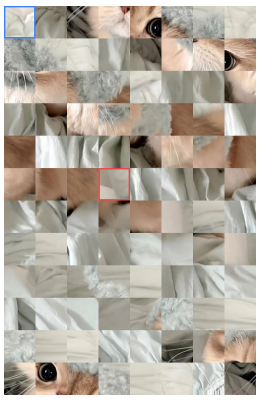
- **Pixel Permutation** rearranges pixel positions without changing their values. In GRT, a flexible transformation matrix is used to disrupt spatial correlations between neighboring pixels [1].
- **Block-Based Scrambling** is applied in our project by splitting the image into several blocks or tiles and then permuting their positions. The original image can be reconstructed by restoring the blocks to their original positions.
- **Reversibility** allows the reconstruction of the original image, while **Bijective Mapping** ensures a "one-to-one" correspondence between original and transformed positions.



(a) SRT: Tile $(0, 0)$ (red box) remains frozen at the top-left corner.



(b) Original unencrypted source image.



(c) GRT: $(0, 0)$ is displaced to (red box); a new tile (blue box) occupies $(0, 0)$.

Figure 2: Visual comparison of origin displacement between (a) Special Rectangular Transform (SRT), (b) the original image, and (c) Generalized Rectangular Transform (GRT).

References

[1] J. Lan and X. Wang, "Generalized Rectangular Transform and Its Application in Image Encryption," *IEEE Access*, 2025.

[2] V. I. Arnold and A. Avez, *Ergodic Problems of Classical Mechanics*. New York, NY, USA: Benjamin, 1968.